

Program Update Day Wednesday 6 February 2019		UU Supervisor
<u>Location: BOL - 1.022: https://students.uu.nl/bolognalaan-101</u>		
13.15- 13.30	<i>Welcome</i>	
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13.30 - 13.45:	Anne Reuten	Krista
13.45 - 14.00:	Jan Schulpen	Nico
14.00 - 14.15:	Eefje Kooij	Nico
14.15 - 14.30	<i>Break</i>	
14.30 - 14.45:	Nathaniel Haanschoten	Iris
14.45 - 15.00:	Joran Westerink	Iris
15.00 - 15.15	Sophia Eijnsman	Stella
15.15 -16.15	Posterpitches + questions at the 11 posters -> max 5 min per poster	
16.30 - 16.45	Rob Koppenaal	Michelle
16.45 - 17.00:	Mila van der Zwaag	Ignace
17.00 - 17.15	Ime Dekker	Estrella
17.15 - 17.30	<i>Break</i>	
17.30 - 17.45:	Jochem Kolsteeg	Annemarie
17.45 - 18.00:	Emma Castellsaguer Petit	Maarten
18.00 - 18.15	Judith Kuijper	Peter
18.15 - 19.00	<i>Drinks</i>	

Poster presentations and UU Supervisor

Anna	Estrella
Bianca	Anouk
Emiel	Marnix
Fabienne	Anouk
Inge	Krista
Joel	Anouk
Joy	Martijn
Layla	Stella
Mats	Annemarie
Mona	David
Piet	Martijn

PPT Presentations

Name Student: Anne Reuten

Title project: Exposing differences in user experience of desktop's and smartphone's websites using eye tracking

Abstract: If you take a moment to look around, what you will see is people behind screens. The abundance of screens and media has transformed humanity into a population of multiscreening skippers and skimmers. This has resulted in competition to create high-value User Experiences (UX). UX is the domain which is occupied with creating satisfying and easy to use digital experiences. UX should however be accommodated to type of device, as each device is inhered with characteristic features. If designers are succeeding, this would result in equal UX for the same websites between devices. This study used eye tracking to investigate potential differences in UX between devices (desktop vs. smartphone) for three Dutch websites. Stimuli consisted of six scenarios and a questionnaire. Forty Dutch students ($M= 22$ years, $SD= 3.73$ years, 29 females) participated. Three metrics were used to benchmark UX: *efficiency*, *effectiveness* and *satisfaction*. Results showed no differences in UX between the desktop's and smartphone's website existed regarding [NS.nl](#) and [Thuisbezorgd.nl](#). Conversely, for [Bol.com](#) a different result emerged, and it was concluded the desktop's website has a somewhat better UX than the smartphone's website. Other findings indicate that against expectations, the number of previous encounters participants had with the website, did not influence UX. Concerning search tools as the filter and sort by button, a potentially exciting idea to classify them according to fixation score (visibility) and amount of participants subsequently deciding *not* to use the tool (judged value), was introduced. Overall, results from this study indicate that designers are quite successful in creating equal UX for different devices, but that improvements are possible.

Location project: Bloakes

External supervisor: Guus van Loon

Name: Eefje Kooij

Title Project: Advisory report to improve job satisfaction at the office of Valori.

Abstract: Since October 2017 Valori B.V. is located in a new office. This new office is an open-plan office. The intention from the management for this new office was mainly to optimize effective interaction and communication between employees. However, there were complaints from the employees, particularly with regard to the high noise level and the degree of distraction caused by colleagues. For the welfare and productivity of the employees, it is important to identify the physical characteristics of the office employees are satisfied with and where there is room for improvement. This was done on the basis of a questionnaire, prepared from literature combined with visions of the management. It measured five dimensions: privacy and acoustics, communication, office features, ventilation and lighting. Based on the results it can be concluded that employees are particularly dissatisfied with the privacy and acoustics within the office. In order to be able to optimize the job satisfaction of the employees alternatives are proposed, on the basis of current research, to improve privacy and acoustics. Alternatives for a greater ability to concentrate are space-in-space spaces, the placing of sound-proof glass walls or concentration corners. In addition, it seems important to invest in alternatives to lower the noise level at the office. Alternatives to this are call chairs, telephone boxes or telephone caps. Due to dissatisfaction in the area of privacy, it might be important to offer employees a space where they can easily have a confidential conversation. This can be done, for example, in a space-in-space space or when there is a telephone conversation using telephone booths. Given the dissatisfaction in the field of visual privacy, an alternative is to place (soundproof) front panels on desks. The concentration corner also offers a workplace with a high degree of privacy.

Location: Valori B.V.

External supervisor: Ria van der Spek

UU supervisor: David Terburg

Name Student: Emma Castellsaguer Petit

Title project: The effect of immersion and active navigation on object recall in virtual reality

Abstract: Virtual environments could be the next step in our search for better tools to help us memorize and recall information. This study answers the question if virtual reality is a good next step in improving education and training, by looking at different concepts: levels of immersion, context-dependent memory, active navigation, and memory schemas. Participants walked around in a virtual office room or looked at a 2D representation of this room. In the room were consistent and inconsistent objects that the participant was told to remember. The results indicate that there was no difference between virtual reality and 2D, but there was a difference between

consistent and inconsistent objects. The inconsistent objects were better remembered and were given a higher confidence score. Although there was no overall difference, higher false alarm rates were found in the 2D condition, these resulted in a higher tendency to answer "Yes, the object was in the room".

Location project: Capgemini

External supervisor: Peter Paul Tonen

UU Supervisor: Maarten van der Smag

Name Student: Ime Dekker

Title project: The impact of images of clowns interacting with children on the amount of a donation for the charity website CliniClowns

Abstract: The website of the CliniClowns uses images to persuade people to donate. Clowns play a substantial role in the advertisements and the behaviour of the clown shown on the images could be a factor influencing the amount of a donation. Witnessing a clown interacting with a child could increase the amount of money in a given donation caused by a feeling of elevation, which is a positive feeling caused by seeing another person helping another. Therefore the aim of this study is whether images of clowns interacting with children would elicit higher donations compared to images of posing clowns for the non-profit organisation the CliniClowns? The results found that images of interacting clowns in comparison with images of posing clowns have a significant positive impact on the amount of a donation. However, a significant result was only found for men and not for women. Elevation could still be a factor in the decision on the amount of money to donate, but there are possibly other factors influencing this decision. More research has to be done for the case of the CliniClowns to examine these other influencing factors and the differences found for men and women.

Location project: Onstuimig

External supervisor: Laurens van Klaveren

UU Supervisor: Estrella Montoya

Name Student: Jan Schulpen

Title project: Evaluating user experiences

Abstract: Advise (a web-developer) wants to gain more hospitals as their customers. They asked me what the factors are for a hospital website that make it a good website. To formulate an answer, I have made a benchmark test of the websites of the five biggest hospitals in the Netherlands.

Furthermore, they also get assignments from costumers who want to make them a webshop. These customers often want copies from websites like [Bol.com](https://www.bol.com), the country's biggest online retailer. But is [Bol.com](https://www.bol.com) a perfect website? To answer this, I have ran user tests to identify the good and bad parts of [Bol.com](https://www.bol.com), so we can give our customers advice on what and what not to copy.

Location project: Advise, Almelo

External supervisor: Tom de Haas

UU Supervisor: Nico Romeijn

Name Student: Jochem Kolsteeg

Abstract: During my internship at CYB I was given two tasks: Teaching and searching images for a new game. There were no clear guidelines on what a 'good' image should look like. During my internship I learned a lot about imagery and communication. I concluded my internship by delivering a set of guidelines for CYB.

Location Project: Cross Your Borders, Nijmegen

External supervisor: Esmee Keijzer

UU supervisor: Annemarie Levert

Name Student: Joran Westerink

Title project: The use of a consumer-grade EEG-device to measure mental workload: A literature study and pilot experiment.

Abstract: The Netherlands Aerospace centre does research on many aspects of aerospace. One of these aspects is the mental workload of individuals working in this sector. An aspect of the operator state is the mental workload. It could be useful to measure mental workload for several reasons: to minimize mistakes; optimize performance of operators, trainings and systems and; to investigate reactions of operators in extreme situations. Physiological measurements of mental workload are useful for in applied research because they can be measured continuously, directly and independent of objective and subjective performance. One such physiological measurement is electroencephalography. The current study focuses on the following questions: Is

measuring electroencephalography a reliable method to measure mental workload in an applied setting and which measurements of electroencephalography are the most reliable? These questions will be answered in two parts: The first part is a literature study on the use of electroencephalography to measure mental workload, including in situations similar to the Netherlands Aerospace Centre. The second part is a pilot experiment to explore the use of electroencephalography to measure mental workload in applied research. The literature study showed that mental workload can be measured using various measurements of electroencephalography, especially power spectral density (Theta, Alpha and Cognitive Load Index) and in various contexts. This includes studies doing research in a similar situation as the Netherlands Aerospace Centre, in more extreme situations, and using lower-grade electroencephalography-devices.

Based on these results, the following experiment was conducted: Eleven participants performed five different difficulties of the NASA Multi Attribute Task Battery II, which is a task designed to manipulate mental workload and to resemble tasks performed by pilots. While performing the task, electroencephalography-signals were measured. The used electroencephalography-device is the consumer-grade Neurosky Mindwave which uses a single, dry electrode. Power values of Theta, Alpha, the cognitive load index and subjective data were compared between the conditions to test if these measurements can distinguish mental workload.

The results showed that power values did not differ significantly between the conditions; No significant differences in mental workload were perceived between the conditions of the task based on the subjective measurements; no correlation was found between the perceived difficulty of the task and the power values.

So, based on the current experiment it could not be concluded that the Neurosky Mindwave electroencephalography-device was able to measure mental workload. This is in contrast with the literature study. The most important explanation is that the Neurosky Mindwave is not able to measure mental workload reliably because of the single electrode and the lack of control in data processing. A second explanation could be the setting of the experiment, which could have led to a high amount of artifacts in the data. The recommendation of the current report is that more research should be done using another EEG-device, data-processing method since it is very likely that EEG could be a reliable method in applied research based on the literature study, despite of the results of the pilot experiment.

Location project: NLR (Netherlands Aerospace Centre)

External supervisor: Maykel van Miltenburg

UU Supervisor: Iris Schutte

Name Student: Judith Kuijper

Title project: Is less really more? A comparative research on the camera presentation for remote bridge operation. (internship)

Abstract: **Introduction:** Current research investigated CCTV-presentation in remote bridge operation. The current camera presentation plan with four alternating images was compared with a new camera plan of six fixed images. **Method:** For this comparison 28 experienced bridge operators executed four bridge operations, two with both camera plans. The camera plans were compared objectively based on the performance of bridge operators using an eye tracker. The eye tracker determined whether the bridge operators viewed the mandatory areas of focus and detected all events. In addition, subjective assessment was taken. **Results:** The results turned out not unambiguous. Objectively, only for one bridge the alternating camera plan performed better. However, this result does not outweigh the objections. The subjective assessment showed a clear preference for the fixed camera plan, supported with valid arguments. Some mandatory areas of focus turned out to be missing on the alternating camera plan. **Interpretation:** This means advice is to replace the current camera plan with the fixed camera plan, which shows all CCTV-images. Recommended future research should be done to justify this advice.

Location project: Vhp Human Performance

External supervisor: Jorrit Jansen

UU Supervisor: Peter Bos

Name Student: Mila van der Zwaag

Title project: Educate-it: Versterk je onderwijs met Blended Learning en een goede website

Abstract: Educate-it is a program of Utrecht University through which the university hopes to keep the level of education high. Educate-it offers different workshops, trainings and tools meant for teachers to improve their teaching skills. Educate-it recently developed a new page on their website to show their selection of workshops, trainings and tools. Previously, this page was found to be very unclear and frustrating for the users to use. On their new page, Educate-it made use of a radial menu to make it easier for the teachers to navigate through the page. By using this menu, their goal was to show their supplies in a clear and structured way.

My goal was to check whether the goals of Educate-it were met by using the radial menu and to make suggestions for better use in the future. I did so by doing usability research on the website. I used eye-tracking to be able to show footage of the participants to Educate-it to give examples of the found problems on the website. I made quite a few recommendations towards Educate-it so that they can develop a better way to show their supplies on their website.

Location project: Educate-it

External supervisor: Suzanne van Hoogstraten

UU Supervisor: Ignace Hooge

Name Student: Nathaniël Haanschoten

Title project: 'Increasing environmental susceptibility by neurofeedback'

Abstract: Humans possess limited mental resources and using Electroencephalography (EEG), demands on attention or workload can be observed in the P300 Event Related Potential component (ERP). One of the components in the P300 is the fP3. The current hypothesis is, that it reflects an inhibition mechanism that frees space to redirect attention towards unsuspected but potentially important events. Expected was that this fP3 ERP could be trained to become larger over time. In the present experiment a neurofeedback condition is compared against a sham feedback control condition over 3 periods of time. In a within-subjects design 4 participants took part in all conditions, that is 'Treatment' with 2 levels (sham / neurofeedback), 'feedback condition' with 2 levels (pre-test / post-test) and 'timebin' with 3 levels (early, midway and late). Mean neurofeedback amplitudes as compared to mean sham amplitudes were not significant, however neurofeedback amplitudes slightly increased over time. Possible causes are the small number of participants, the short period of training and the time between training sessions. Opportunities for future research are testing a very short period of training, as was the case in the present experiment, against a much longer training period and observe if this makes any difference, using more participants or combine the two options to examine possible interactions. Although it is likely that a fP3 training will be found and developed for application to maintain environmental susceptibility in automated driving, it is very important that such a training doesn't affect reaction times for a primary task. This seems to be another direction for future research.

Keywords: Electroencephalography, Event Related Potentials, susceptibility, applied cognition

Location project: Langeveld Building, Utrecht University

External supervisor: -

UU Supervisor: dr. Iris Schutte

Name Student: Rob Koppenaal

Title Project: UX Researcher

Abstract: During this internship, I have led three projects. During my presentation I will talk about one project. Furthermore, I share my experience of working in a corporate environment.

The project I will talk about: a problem that could arise if employers are not aware of each other's wisdom is that they do similar research that should only be researched ones. To prevent this, a system is designed where people can upload rapports and share their insights. I wrote a rapport that combines various studies that cross the topic of onboarding processes. Onboarding processes are procedures that people experience when they use a system for the first time. This rapport will save a lot of time and energy for people that previously had to read and compare all the various studies.

Location project: ABN AMRO

External supervisor: Suzanne Vosslander

UU supervisor: Michelle Emons

Name Student: Sophia Eijssman

Title project: Validation on the EmojiGrid

Abstract: Since the relation between emotions and olfaction is one that has yet to be carefully understood, there is need for a cultural and age independent assessment tool that measures smell related emotions. In a previous paper, the EmojiGrid has been introduced as a tool that is validated to measure food-related emotions with visual stimuli. The EmojiGrid is a grid that has been labeled with 16 different emotions (emoji) and measures the self-report of the affective state. This means that the users of the EmojiGrid are able to report their affective state, valence and arousal, within the EmojiGrid towards the stimuli that are being presented. Since the EmojiGrid has already been validated for visual stimuli, it is important for future studies in olfaction that it can be validated for

olfactory stimuli (smells) as well. Here, we hypothesize that the EmojiGrid is a valid tool to assess valence and arousal for smells. To test whether the EmojiGrid is a valid tool to assess valence and arousal for smells, we performed an experiment in which the participants used the EmojiGrid to rate the valence and arousal for 40 different smells. The smells varied from food to non-food products, from flowers to faeces and therefore covers a large range of the smell affective state.

Location project: Langeveld gebouw, Uithof

External supervisor: Lex Toet

UU Supervisor: Stella Donker

Poster Presentations

Name Student: Anna Hoonakker

Title project: UX research in a digital commercial environment

Abstract: This working internship was executed at a company whose products and services rely heavily on user experience and UX research. Here, I worked together with UX research experts and validated newly designed or iterations of user interfaces. My main goal coming in to this internship was gaining useful experience and knowledge in the field of UX research, but also getting familiar with the eye tracker as a usability tool. Daily activities consisted of attending meetings, getting to know the company and its processes, taking notes during research days, interviewing customers about ideas, products and interfaces, writing reports on how customers experience products and interfaces and include advice on how to improve these products and interfaces. Except for getting familiar with the eye tracker and its software, I also validated an icon set, which is used in all the company's online interfaces. Results from this validation show not all icons being understood by the customers. By carrying out this project I created a template which can be used by employees for further validating more icons.

Location project: ABN-AMRO

External supervisor: Ike Breed

UU Supervisor: Estrella Montoya

Name Student: Bianca Beerepoot

Title project: Evaluative feedback for a VR job interview training

Abstract: Having good conversation skills is crucial during a job interview and in life. You can really make changes if your skills are well developed, therefore, training them is very important. During the learning process feedback is the key to success. Feedback should be evaluative by informing the trainee what their goal is, how they are doing and where they have to go next to achieve their goal. This is implemented in a (not yet launched) job interview training created for CoVince. The training still needs some adjustments, but has potential to become the guide for many people!

Location project: CoVince

External supervisor: Richard van Tilborg

UU Supervisor: Anouk Keizer

Name Student: Emiel Cuijpers

Title project: Presence: the Sense of 'Being There' in Immersive Virtual Reality

Abstract: Most VR designers have been engineers with the expertise in technology, but limited in their understanding of human perception and cognition. However, a basic understanding how we perceive reality will place emphasis on the human side in the communication between human and machine, which, in turn, will contribute to the design of innovative and intuitive virtual environments. Achieving presence – the sense of 'being there' in the virtual environment - is a frequently emphasized goal when discussing the design of virtual environments and this objective affects both the academic and applied field of work. The goal of this literature study is to provide an overview, which perceptual processes contribute to the forming of presence in an immersive virtual environment. This review will also provide the most prominent immersive factors of a VR system that contribute to the feeling of presence. Results show that VR designers, concerned with a fixed financial budget, are advised to place emphasis on the tracking level, stereoscopy and field-of-view of the system rather than visual realism or auditory quality to create a sense of presence in the virtual environment. Additionally, it was found that a corresponding gaze direction from the virtual characters to the participant's movement and environmental sound (e.g. birdsong) that matched the outdoor of the virtual room were important factors that contribute to the illusion that the events in the virtual environment are real.

Location project: VR Lab, Nijmegen
External supervisor: Leon Phillich
UU Supervisor: Marnix Naber

Name Student: Fabienne Heijn

Title project: A Conversation Model For Virtual Job Application Training

Abstract: There are multiple ways to train structured conversations in different situations. A first date, a job application or even a bad news conversation are all examples of such kind. These conversations can be trained in real life, but this can get expensive, hard to organize or maybe even considered awkward for the persons involved. By making use of virtual reality, another possibility to train these kinds of conversations is created. With the help of a conversation model, people can be trained based on a specific set of learning goals while it allows the system to respond appropriately to the user's answers. This way, the training can be structured in such a way that it will gradually work towards the pre-set goals, while keeping the trainee's answers in mind, resulting in a smooth and natural conversation.

Location project: CoVince - Adventurous Learning

External supervisor: Richard van Tilborg

UU Supervisor: Anouk Keizer

Name Student: Inge Koehorst

Title project: Probabilities of the Conceptual Priming Test

Abstract: Bloakes has created an Implicit Reaction Time Test, called a Conceptual Priming Test. The current study examined if the Conceptual Priming Test is suitable to measure the influence of price on the perceived quality. In this test, response times were measured for categorizing positive and negative attributes after being primed with a target that showed a product (battery packs from Albert Heijn or Duracell) that was priced with 0.59, 3.49, 6.99 or 15.29. Because of the *price - perceived quality* relationship we expected that participants would take a long time to categorize positive attributes and a short time to categorize positive attributes when a product is priced way too low. We expected the same when a product is priced way too high, because the sacrifice the participant has to make to purchase the product is too big. The results show that participants were significantly slower in categorizing *favoriet* (favorite) after seeing the Albert-Heijn batteries combined with the prices 0.59 and 15.29 than when the batteries were combined with 3.49, which means the participants had a more positive association with the batteries when they were combined with their actual price. For the Duracell batteries it applies that participants were significantly faster in categorizing *slecht* (bad) after seeing these batteries combined with the price 0.59 than after seeing the Duracell batteries combined with the other three prices, which implies a more negative association with the batteries when they are priced 0.59. Lastly, recommendations for further research will be discussed.

Location project: Bloakes, Waardenburg

External supervisor: Guus van Loon

UU Supervisor: Krista Overvliet

Name Student: Joël Tan

Title project: Rating and measuring perceived quality in oral presentations: using a comprehensive multimodal tool

Abstract: This study consists of an exploratory research into CoVince Adventurous Learning. CoVince is a company based in Utrecht which developed a virtual reality application for training a variety of skills. In this study however, only their presentation trainer has been included. The presentation trainer CoVince developed can be used on any device, with the option of using virtual reality. This makes it possible to practice an elevator pitch to Donald Trump in an actual elevator or to present in front of a full theater. After these training sessions, users receive automated feedback, which brings us to the current study. Until now, CoVince has not studied when to give a certain feedback message. In order to find out, quantified measures need to be formed in order to give objective automated feedback, based on how people rate presentations. This study explores a few verbal measures for the CoVince presentation trainer.

Location project: Utrecht

External supervisor: Richard van Tilborg

UU Supervisor: Anouk Keizer

Name Student: Joy Zwart

Title project: Research and consulting on medical device usability

Abstract: UL is a global safety consulting and certification company. Their office in Utrecht is focused on Human Factors engineering of medical devices. They collaborate with medical device manufacturers to help make their products safe, effective, appealing and easy to use. In contrast to most internships, I didn't work on an individual project. I was involved in several new and ongoing projects and worked on them in different teams. My main tasks were; conducting user research, conducting usability tests and developing training materials for Human Factor related trainings. On my poster I will show an overview of the Human Factors Engineering process of developing a safe and easy to use device, accompanied by some examples of projects I've worked on.

Location project: Emergo by UL

External supervisor: Veerle Migchelbrink

UU Supervisor: Martijn Mulder

Name Student: Yingxuan Liu (Layla)

Title project: The Influence of Smell Detection Threshold on Odor-Evoked Affects and the Perceived Intensity toward Odors

Abstract: Smells can cause the arousal of affective valence at different levels, while individual olfactory ability should be taken into account as a reference for the perceived intensity of scents and the degree of valence and arousal. This study tested the correlations between valence and arousal, the perceived intensity, and smell detection threshold in term of unpleasant and pleasant olfactory stimuli. There were 56 people participating in our study. The results showed that for both unpleasant and pleasant olfactory stimuli people experienced less arousal for more positive affect. Perceived intensity was not correlated with either valence and arousal. Generally, there was no correlation between threshold and valence, arousal, or perceived intensity respectively. But in the female group, for the unpleasant stimuli, the more sensitive they are toward smells, the more negative affect they experienced. Limitations and implications were discussed.

Location project: TNO

External supervisor: Lex Toet

UU Supervisor: Stella Donker

Name Student: Mats Bakker

Title project: Why do physiotherapists not look at what they want?

Abstract: SilverFit develops exergames that help elderly with the rehabilitation from physical and psychological injuries. The mole game provides clinimetric data of the patient after the game is played. Clinimetric data shows different outcome measures which show the objective progress that patients make. Despite the fact that physiotherapists find clinimetric data interesting, clinimetric data on the SilverFit 3D is not often used. The purpose of this study therefore is to find out why physiotherapists do not use the clinimetric data on the SilverFit 3D. In this study four physiotherapists were observed and nine physiotherapists were interviewed on using the clinimetric data after the mole game. To ensure the physiotherapists would behave as normal as possible, the physiotherapists did not know that they were observed. After the observations a semi-structured interview was conducted. The results show that the clinimetric data cannot be compared over time, is in some cases unclear, is not always easy to understand and physiotherapists have to perform many actions before all clinimetric data can be seen. A new output screen for the clinimetric data is designed that takes these restrictions into account. This could help to increase the usage of the clinimetric data on the SilverFit 3D. In the future more exergames on the SilverFit 3D will be provided with clinimetric data, and contribute to increasing usage of clinimetric data on the Silverfit 3D.

Location project: SilverFit B.V. Woerden

External supervisor: Kevin Verleijdsdonk

UU Supervisor: Annemarie Levert

Name Student: Mona Zimmermann

Title project: Inclusive design for everyone: Developing content for the DID-IT toolkit

Abstract: TNO is working on an "inclusive design toolkit" for people involved in the development of digital services (for the government) in order to encourage designs that are usable for all citizens. Inclusive design is the process of designing with and for (vulnerable) users. TNO strives to make inclusive design more tangible for the stakeholders involved, by providing information on the Why?, What? And How? of inclusive design. During the here presented internship I worked on developing content and a way of presenting that content as part of this toolkit. Specifically, I researched information about psychological characteristics of low-literates that need to be accommodated for when designing digital services in an inclusive way. Also, design guidelines for the design

with and for low-literates were researched. Interviews were conducted to understand how stakeholders would like the content to be structured and presented in the toolkit. The results led to a preliminary design of the toolkit that was accompanied by posters that should act as triggers to design inclusively with the inclusive design toolkit. As a final step of the internship, 3 poster prototypes were designed that present some of the content that was developed in the first part of this internship.

Location project: TNO Soesterberg

External supervisor: Rosie Paulissen

UU Supervisor: David Terburg

Name Student: Piet Jonker

Title project: Decision making of maritime pilots on the bridge of a ship versus the bridge of a maneuvering-simulator

Abstract: Maritime pilots have the complex task to navigate ships into the dock. To learn to handle large ships (350+ meters), maritime pilots receive simulator training. In this research, decision-making of pilots during dock navigation was mapped and compared between the simulator and on board. Our participant is a cooperating pilot from the port of Rotterdam. He was observed on board with cameras, audio recorders, eye tracking and ECG. Later, the journey was replicated in the simulator where identical observations were made. Observations have been divided into blocks of subtasks. This resulted in the identification of seven blocks and a system analysis. Comparisons of on board versus simulator yielded various relevant differences. In conclusion, the journey could be divided into blocks of subtask, where decision-making could be investigated. In the simulator, less visual relevant information was present, resulting in more dependency on the equipment.

Location project: MARIN (Wageningen)

External supervisor: Wendie Uitterhoeve

UU Supervisor: Martijn Mulder